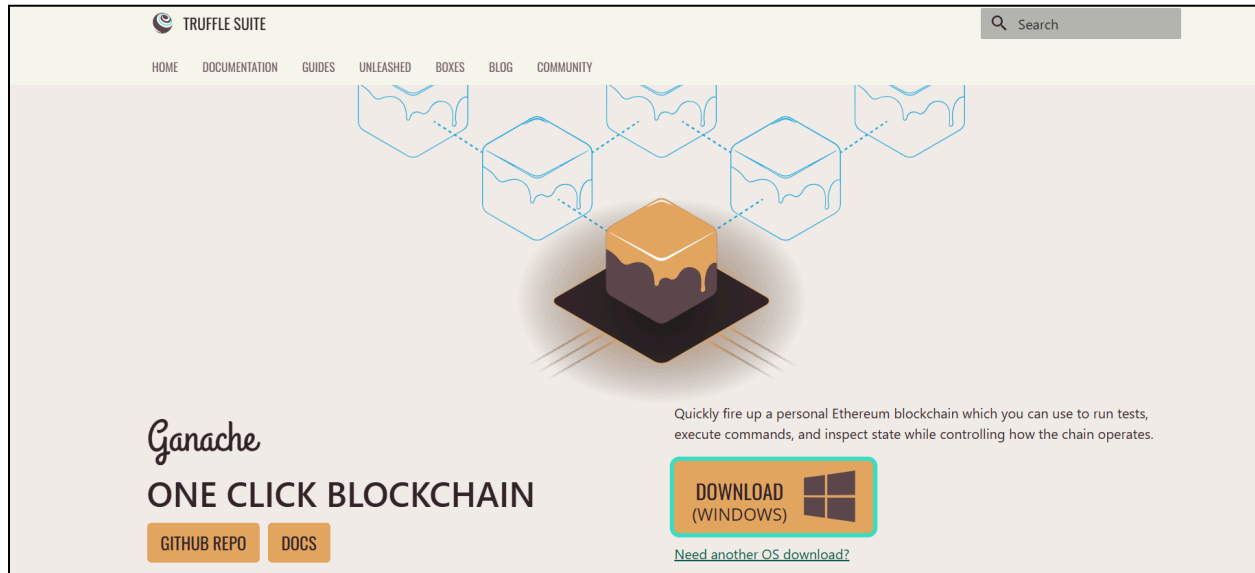


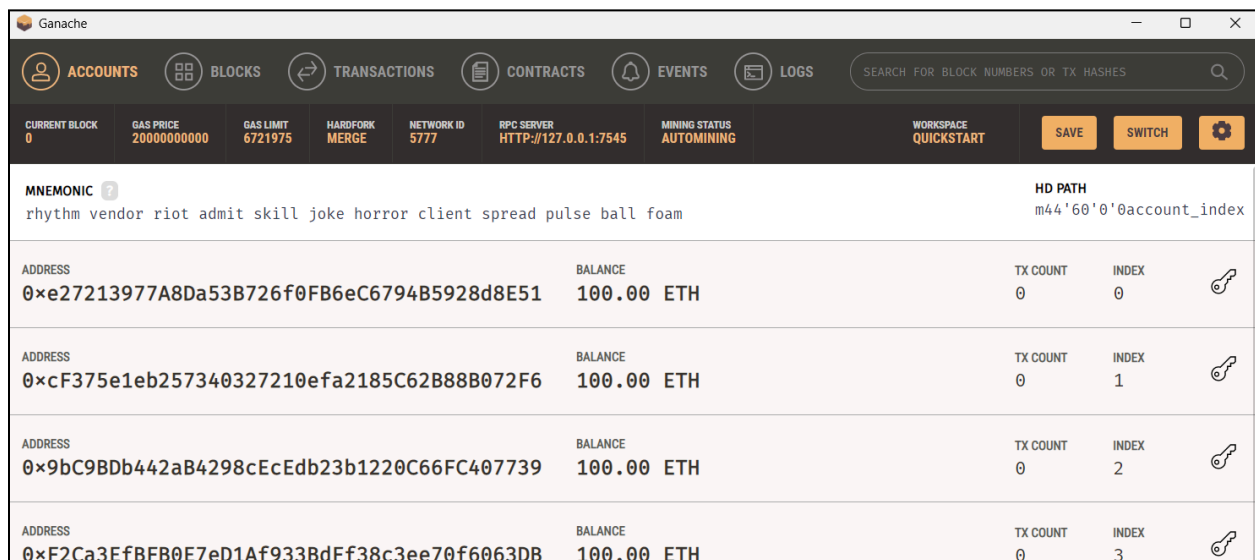
Experiment 8

Aim: Implement the Blockchain platform Ganache

Step 1: Download and install ganache



Once installed, open ganache and click on Quickstart Ethereum to start localhost.



Step 2: In MetaMask, click on the networks tab and add a custom network.

< Ganache Local X

Network name

localhost 8545

Default RPC URL

ganache
127.0.0.1:7545

Chain ID

1337

Currency symbol

ETH

Block explorer URL

Add a URL

Save

Step 3: Import ganache account into metamask

< Add wallet

Private key

Imported private keys are backed up to your account and sync automatically when you sign in with the same Google or Apple login.
[Learn more](#) about how imported keys work

Select type

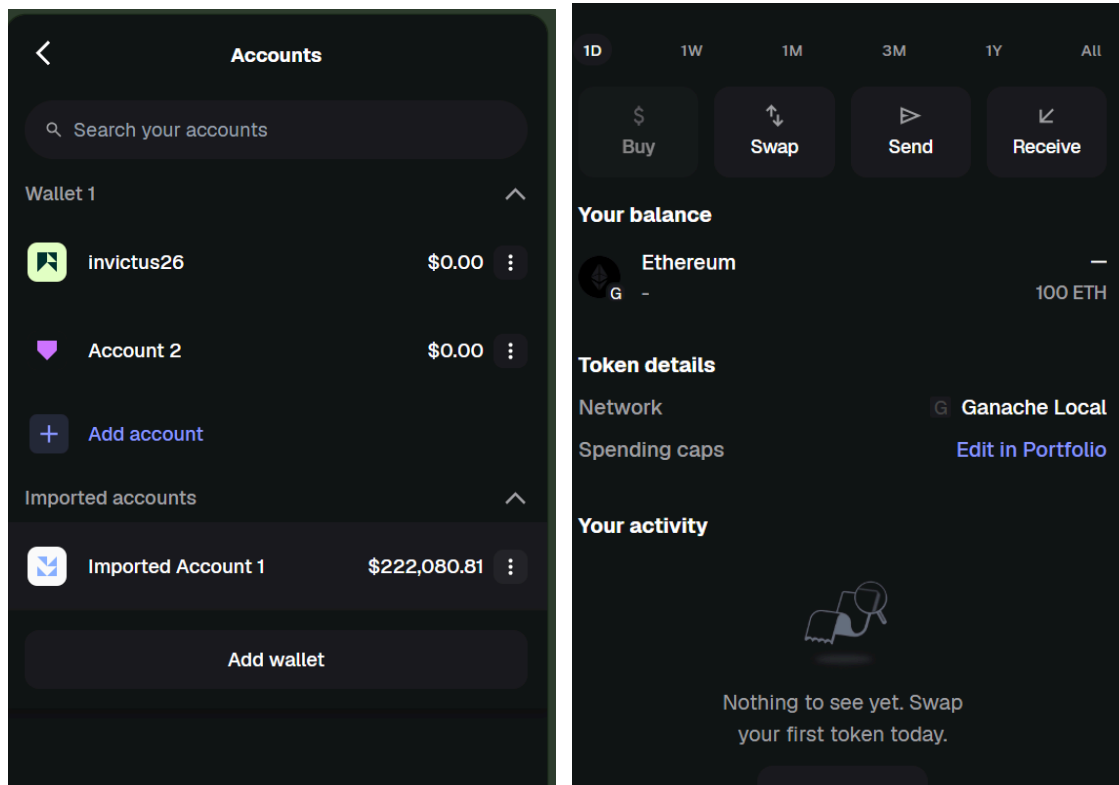
Private key

Enter your private key string here:

.....

Cancel Import

Wallet balance appears in metaMask account



Step 4: In Remix IDE, create a new file **crowdfunding.sol**

Code:

```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.0;
```

```
contract Crowdfunding {  
    address public owner;  
    uint public goal;  
    uint public deadline;  
    uint public totalFunds;
```

```
    mapping(address => uint) public contributions;
```

```
    constructor(uint _goal, uint _duration) {  
        owner = msg.sender;  
        goal = _goal;
```

```

        deadline = block.timestamp + _duration;
    }

    function contribute() public payable {
        require(block.timestamp < deadline, "Deadline passed");
        require(msg.value > 0, "Send some ETH");

        contributions[msg.sender] += msg.value;
        totalFunds += msg.value;
    }

    function withdrawFunds() public {
        require(msg.sender == owner, "Only owner");
        require(totalFunds >= goal, "Goal not reached");

        (bool success, ) = payable(owner).call{value: totalFunds}("");
        require(success, "Transfer failed");
    }

    function refund() public {
        require(block.timestamp > deadline, "Not ended");
        require(totalFunds < goal, "Goal was reached");

        uint amount = contributions[msg.sender];
        require(amount > 0, "No contribution");

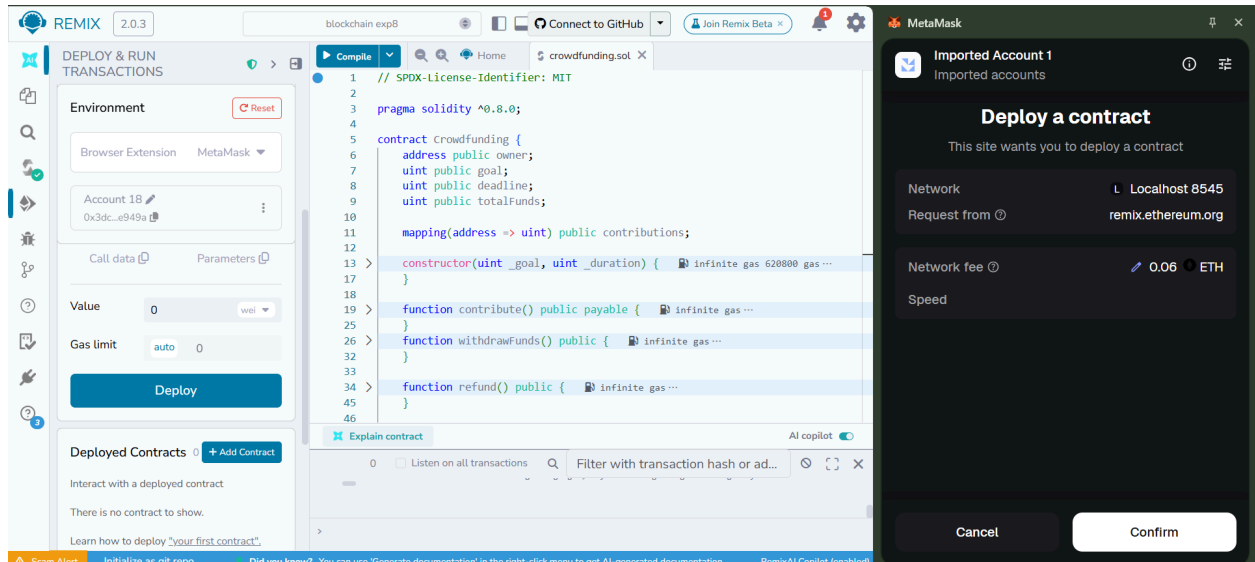
        contributions[msg.sender] = 0;

        (bool success, ) = payable(msg.sender).call{value: amount}("");
        require(success, "Refund failed");
    }

    function getBalance() public view returns (uint) {
        return address(this).balance;
    }
}

```

Step 5: Compile and run the contract.
 When prompted in metamask, confirm the transaction.



The contract is deployed!

The deducted amount can be seen in ganache app along with transaction count

Ganache					
ACCOUNTS BLOCKS TRANSACTIONS CONTRACTS EVENTS LOGS					
SEARCH FOR BLOCK NUMBERS OR TX HASHES					
CURRENT BLOCK 2	GAS PRICE 20000000000	GAS LIMIT 6721975	HARDFORK MERGE	NETWORK ID 5777	RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING				WORKSPACE QUICKSTART	SAVE SWITCH
MNEMONIC apology essence feed north arm announce receive axis total wall chase involve					
HD PATH m44'60'0'8account_index					
ADDRESS 0x678aB28b947030404778C6193423B73c7a014b18	BALANCE 99.92 ETH	TX COUNT 2	INDEX 0		
ADDRESS 0xcC163421b1091647e96D5Ce9ddEdF1e624dad0a6	BALANCE 100.00 ETH	TX COUNT 0	INDEX 1		
ADDRESS 0x7Ea6E7307F61C1b29eA2FC7c40F16e602b514a31	BALANCE 100.00 ETH	TX COUNT 0	INDEX 2		
ADDRESS 0xb00C1B0A9e7E90b2b538A98d7DD98A499578F4dd	BALANCE 100.00 ETH	TX COUNT 0	INDEX 3		
ADDRESS 0x176239f7D57783b2Fe396cb7FA3B73178C750713	BALANCE 100.00 ETH	TX COUNT 0	INDEX 4		
ADDRESS 0x9BFCFd4b9223c4d5d551bb8152DBAFFAF449A36B	BALANCE 100.00 ETH	TX COUNT 0	INDEX 5		

